

Federico Alberto Astolfi

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Research Interests

Quantum optimal control · Quantum error correction · Machine learning for quantum information · Neutral-atom quantum computing · Rydberg systems · Pontryagin Maximum Principle

Education

Oct 2024 – present

Ph.D. in Physics

Université de Strasbourg, CESQ & ISIS, Strasbourg, France

- Supervisor: Prof. Guido Pupillo.
- Funded by the Marie Skłodowska-Curie Actions (MSCA) doctoral network *Machine Learning for Quantum* (MLAQ).
- Research areas: Quantum Optimal Control (Pontryagin Maximum Principle, GRAPE) applied to neutral-atom/Rydberg systems; Machine Learning approaches for decoding Quantum Error Correcting Codes.

Sep 2021 – Feb 2024

M.Sc. in Mathematics

Università degli Studi di Milano, Milan, Italy

- Final grade: 110/110 *cum laude*.
- Thesis: *Minimal Lagrangian orbits of Hamiltonian actions over KE and KRS*.
Abstract. Hamiltonian actions of a Lie group on a symplectic manifold admit a moment map from the manifold to the dual of the Lie algebra of the acting group. We give an explicit formula for the moment map when the manifold is a Kähler–Ricci Soliton, then study minimality of Lagrangian orbits for semisimple acting groups. Under certain hypotheses, there exists a unique Lagrangian orbit, which is moreover minimal.
- Erasmus exchange (6 months) at Rheinische Friedrich Wilhelms Universität Bonn, Germany.

Sep 2018 – Jul 2021

B.Sc. in Mathematics

Università degli Studi di Milano-Bicocca, Milan, Italy

- Final grade: 110/110 *cum laude*.
- Thesis: *Una introduzione alla teoria di Morse*.

Sep 2013 – Jun 2018

Diploma in Classical Studies

Liceo Classico Giulio Casiraghi, Cinisello Balsamo, Italy

- Final grade: 100/100.

Publications

- [1] **F. A. Astolfi**, S. Jandura, G. Pupillo, “Pontryagin Maximum Principle for Rydberg-blockaded state-to-state transfers: A semi-analytic approach,” *Phys. Rev. Research* **8**, 023162 (2026). DOI:10.1103/2w2v-hpst arXiv:2512.13549

Research & Professional Experience

Oct 2024 – present	Ph.D. Researcher <i>Centre Européen des Sciences Quantiques (CESQ) & ISIS, Strasbourg, France</i> - Developed a semi-analytic optimal control framework for Rydberg-blockaded two-qubit gates, combining Pontryagin Maximum Principle with numerical optimisation (GRAPE). - Investigated ML-based decoders for topological quantum error correcting codes (surface code, toric code). - Participated in the ML4Q network: workshops, secondments, and collaborative research across European institutions.
Apr 2025 – Jun 2025	Research Intern <i>Leithà – Unipol Group, Bologna, Italy</i> Investigated hyperparameter optimisation techniques for deep neural networks in a financial-modelling context.
Dec 2023 – Aug 2024	Test Engineer <i>TXT Group, Cologno Monzese, Italy</i> - Software testing for aero-machinery projects (client: Leonardo Helicopters). - Designed and executed test plans ensuring quality, functionality, and performance of embedded software.

Talks & Conferences

- QIP 2025**, Riga, Latvia — Attended (January 2026).
- ML4Q Network meetings** — Regular participant (2024–present).

Technical Skills

Programming	Python, Julia
Key libraries	NumPy, SciPy, PyTorch, Qiskit, QuTiP
Methods	Optimal Control (PMP, GRAPE), Machine Learning (deep learning, reinforcement learning), Quantum Error Correction
Tools	Git, L ^A T _E X, Overleaf

Languages

Italian	Native speaker
English	Proficient (C1)
French	Lower intermediate (B1)

Grants & Funding

- **MSCA Doctoral Fellowship** – Marie Skłodowska-Curie Actions, Horizon Europe, doctoral network *Machine Learning for Quantum* (Oct 2024 – present).

Other Experience

May 2018 – Dec 2023 **Swimming Teacher & Lifeguard**

A.S.D. Asa, Cinisello Balsamo, Italy

- Coached swimmers of all ages and skill levels; held pool-safety certification.